REJUYAN HOSSAIN SHAWON

Software Engineer | Full-Stack & Applied AI Developer

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PROFESSIONAL SUMMARY

Computer Science M.S. candidate with 16 months of professional software development experience, specializing in full-stack web development, RESTful API integration, database optimization, and performance improvement. Independently designed, developed, and deployed three full-stack applications incorporating secure authentication, data analytics, and AI-powered functionality. Experienced in React, TypeScript, Node.js, Python, and MongoDB, with research expertise in multimodal deep learning, computer vision, and vision-language models. Strong problem-solving skills demonstrated through measurable software performance improvements and research-driven technical solutions.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages: TypeScript, JavaScript, Python, SQL, C++, C# | Frontend: React, Redux, HTML, CSS, Bootstrap, Tailwind CSS
Backend & Data: Node.js, Express.js, REST APIs, FastAPI, MongoDB, MySQL, MS SQL Server | AI: PyTorch, RAG, Gemini API
Tools & Competencies: Git, GitHub, Heroku | Full-Stack Development, API Optimization, Database Design, Problem Solving, Team Collaboration, Code Reviews

PROFESSIONAL EXPERIENCE

Junior Software Developer | Urmee Knitwear Limited | Feb 2022 – May 2023

- Developed and maintained MERN web applications, building frontend features and integrating REST APIs with backend services.
- Analyzed API performance bottlenecks and optimized database queries, reducing API loading time by 30%.
- Built 10+ responsive React UI components, improving frontend performance by 20%.
- Collaborated with developers through code reviews and troubleshooting, contributing to a 15% reduction in bugs.

SELECTED PROJECTS | INDEPENDENTLY DEVELOPED & DEPLOYED

FinTrack — Personal Finance Manager | React, Node.js, Express, MongoDB | Source Code

- Built a MERN personal-finance platform to help users track spending and manage budgets; implemented JWT-secured accounts, transaction CRUD, category budgets, CSV export and MongoDB persistence.

ContextAI — Document-Aware AI Application | MERN, Gemini API, RAG | Source Code

- Developed document-grounded Q&A using PDF/TXT parsing, text chunking, embeddings, cosine-similarity retrieval, Gemini API integration and JWT authentication.

PhonePulse AI — Smartphone Sentiment Analytics | MERN, analytics | Source Code

- Developed smartphone brand/model and feature-level sentiment dashboards using CSV ingestion, rule-based scoring and regional filters with clearly labeled demonstration data.

Live demos: FinTrack | ContextAI | PhonePulse AI

SELECTED RESEARCH

- HGR+++ — Multimodal Fake News Detection: designed metadata-conditioned fusion of CLIP text/image features for six-class classification; achieved 96.59% accuracy and 94.90% macro-F1 on Fakeddit.
- CNN–Transformer Image Captioning: developed an encoder–decoder approach for image caption generation, investigating label smoothing, backbone fine-tuning and length-normalized decoding.

EDUCATION & RELEVANT COURSEWORK

The City College of New York (CUNY) | M.S. Computer Science | Expected May 2027
Coursework: Advanced Algorithms, Database Systems, Computer Vision, Applied Cryptography; Adversarial AI and Neural Networks (in progress).

Daffodil International University | B.Sc. Computer Science & Engineering | May 2018 – May 2022
CGPA: 3.47/4.00 | ECE-evaluated GPA: 3.61/4.00

LEADERSHIP, TRAINING & LANGUAGES

Leadership & Service: Founding Vice President, Madok Nirmul Committee; founding member, Hand for Help; blood donation community volunteer.
Training: MERN Web Development, Programming Hero | Languages: Bangla (native), English (professional proficiency)